

real 3D car User Manual

Three.js 3D racing game play guide

Web game URL: <https://real-3dcar.streamlit.app>

Document version: 1.0

Date: 2026-07-31

1. Game Overview

real 3D car is a 3D racing game where you avoid obstacles, manage speed, and build a high score. You can select a difficulty before starting, steer around incoming obstacles, and earn extra points through the multiplication mini-game during straight road sections.

2. How to Launch

- 1. Open the project folder: `C:\work\AI\LLM\codex\3dCar`
- 2. Start the development server: `npm run dev`
- 3. Open the Vite URL in your browser, usually `http://localhost:5173`.
- 4. For production output, run `npm run build`.

3. Start Screen

Select a language and difficulty, then press **Start Game**.

Label	Meaning
S	Straight road duration
C	Curved road duration
M	How long the mini-game panel is displayed

4. Difficulty Settings

Difficulty	Obstacle Speed	Straight	Curve	Mini-game Display	Score Multiplier
Beginner	18	20 sec	5 sec	5 sec	1.00x

Difficulty	Obstacle Speed	Straight	Curve	Mini-game Display	Score Multiplier
Intermediate	26	20 sec	10 sec	10 sec	1.22x
Advanced	34	10 sec	10 sec	10 sec	1.48x
Pro	44	5 sec	20 sec	5 sec	1.82x

5. Controls

Action	Keyboard	Mobile
Move left	A or Left Arrow	LEFT button
Move right	D or Right Arrow	RIGHT button
Speed up	W or Up Arrow	Keyboard recommended
Slow down	S or Down Arrow	Keyboard recommended
Restart	R	Restart Race button

6. Dashboard

- **Speed:** Current vehicle speed
- **Score:** Current score
- **Distance:** Distance traveled
- **Best Score:** Highest score saved in the browser
- **Difficulty:** Selected difficulty
- **Obstacle Speed:** Base obstacle speed and live speed input change

7. Scoring Rules

- Driving score increases over time based on speed and difficulty.
- Higher driving speed gives a higher score weight.
- Higher difficulty gives a higher score multiplier.
- Correct mini-game answer: **+300 points**.
- Wrong answer or timeout: **-300 points**.
- Score never drops below zero.

8. Multiplication Mini-game

During straight road sections, a multiplication mini-game panel appears on the right side of the screen. Each question allows **3 seconds** for answer input.

- Beginner and Intermediate: the mini-game appears 5 seconds after a straight section starts.
- Advanced and Pro: the mini-game appears immediately when a straight section starts.
- A fanfare plays when the mini-game starts.
- Correct answers show **Nice!** with a bright sound.
- Wrong answers or timeouts show **Not good!** with a short low sound.

The difficulty value controls how long the mini-game panel stays visible. It is separate from the 3-second answer timer.

9. Crash and Restart

If your car collides with an obstacle, the race stops and the crash screen appears. You can select a new difficulty and press **Restart Race** to begin again.

- Mini-game questions and timers are reset after a crash.
- The selected difficulty is reflected in the dashboard.
- Best score is saved in local browser storage.

10. Car Photo Upload

Use **Upload Photo** to place an image on the roof of the car. Use **Clear** to restore the default placeholder.

11. Music and Sound Effects

- Use **BGM On/Off** to toggle background music.
- A crash sound plays on collision.
- Mini-game start, correct answer, and wrong answer each have their own sound effect.

12. Troubleshooting

Issue	What to Check
Game screen is blank	Check that the development server is running, then refresh the browser.
Mobile steering buttons are missing	Clear browser cache and confirm the viewport is mobile sized.
No sound	Check browser tab mute, OS volume, and whether you clicked the page once to enable audio.
Mini-game does not appear	Confirm the road is in a straight section. Beginner and Intermediate wait 5 seconds before showing it.

This manual is based on the local project at `C:\work\AI\LLM\codex\3dCar`.